

Week 5: Simple Linear Regression I

PLSC 309 Quantitative Political Analysis

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Problem Set 2

- Due: **tomorrow (Sep 23rd), 11:59 p.m. (midnight).**
- Submit on Canvas

Feedback: Final Project Checkpoint 1

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- ① **Data availability:** Confirm that your data are accessible and downloadable.
- ② **Conceptual clarity:** Clearly define your key concepts. For example, what do you mean by “foreign policy behaviors” or “disadvantaged communities”?
- ③ **Clear, concise language:** For a research question, prefer “Does education shape voter turnout?” to “Does people’s education level impact their turnout of vote?”
- ④ **Scope of the theory:** Specify the population to which your theory applies. If you study social media and political participation among young people, is your theory specific to young people, or are you testing a general theory using this population?

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Checkpoint 2: Deadline Extended

The new deadline is **October 14, 11:59pm (midnight)**.

Weeks 3-4: Fundamentals of Probability and Statistics

- Expected value: $E(X)$
- Variance: $\text{Var}(X) = E[(X - E[X])^2]$.
- Covariance: $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$.
- Estimand, Estimator, Estimate
- Mean Squared Error: $\text{MSE} = E[(Y - \hat{Y})^2]$.

Today's Agenda

Simple Linear Regression (two-variable linear regression model)

- Conditional Expectation Function (CEF)
- Best Linear Predictor (BLP)
- Ordinary Least Squares (OLS) estimator

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#) and from Brandon Stewart's [SOC 500 course materials at Princeton](#).

1. Conditional Expectation Function

Shapiro Example

Suppose we are campaign analysts predicting voter support for Governor Shapiro.

For each voter, define:

$$Y = \begin{cases} 1 & \text{if the voter supports Shapiro,} \\ 0 & \text{otherwise.} \end{cases}$$

Our first question is:

What proportion of voters supports Shapiro?

Because Y is binary, the population support rate is:

$$\mu = E[Y] = P(Y = 1).$$

Beyond the Overall Average

The overall support rate does not tell us how support differs across age groups.

We now ask:

What proportion of voters within each age group supports Shapiro?

Let X denote a voter's age group.

We want to know:

$$E[Y \mid X = 18-24], \quad E[Y \mid X = 25-31], \quad \dots$$

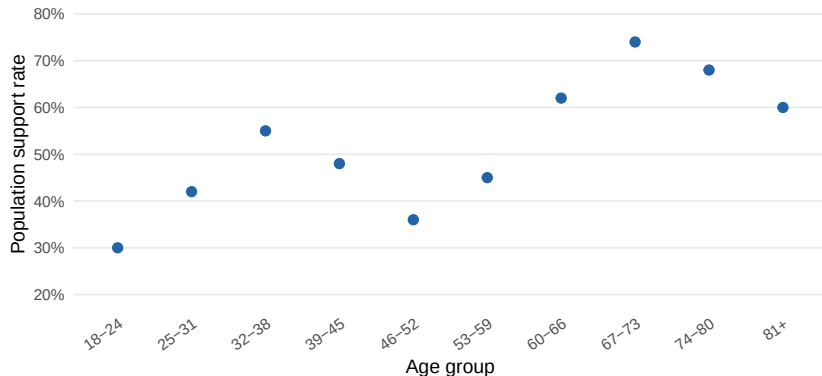
Each quantity describes the average support **within one group**.

An Illustrative Population

Suppose we observe the entire population and calculate the support rate within each age group:

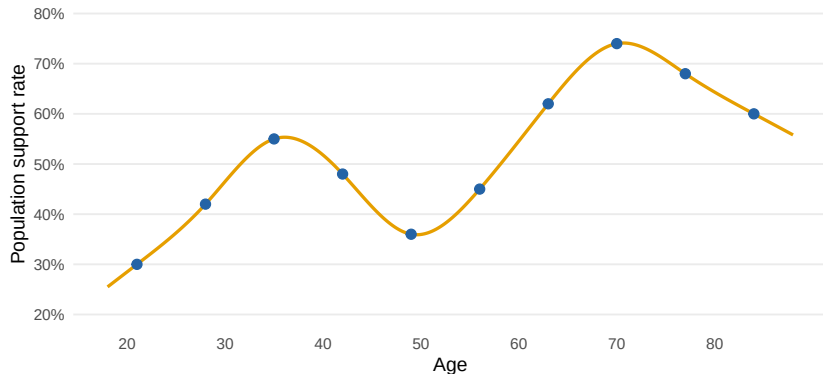
Age group x	Support rate $E[Y X = x]$
18–24	30%
25–31	42%
32–38	55%
39–45	48%
46–52	36%
53–59	45%
60–66	62%
67–73	74%
74–80	68%
81+	60%

Voter Support by Age Group



Each point represents the population support rate within an age group

Voter Support Across Continuous Age



When age is treated as a continuous variable, we can represent the population support rate as a smooth function of age. The smooth curve shows the population support rate at each age.

The Conditional Expectation Function

The **Conditional Expectation Function** (CEF) of Y given X is the function that assigns to each value x the conditional mean of Y :

$$m(x) \equiv E[Y \mid X = x].$$

- $m(x)$ is the mean of Y in the population with $X = x$.
- Evaluating this function at the random variable X gives $m(X) = E[Y \mid X]$.

In our example, X denotes age and Y indicates support for Shapiro. The input x is a specific age; the output $m(x)$ is the population support rate among voters of that age.

$$m(21) = E[Y \mid X = 21] = 0.30,$$

$$m(28) = E[Y \mid X = 28] = 0.42.$$

Thus, 30% of 21-year-old voters and 42% of 28-year-old voters support Shapiro.

From the CEF to a Linear Approximation

- The CEF can be infinitely complex.
- Estimating the CEF can therefore be rather difficult.
- So, instead we can limit ourselves to classes of functions that approximate the CEF.

In particular, what if we limit ourselves to the class of **linear predictors**:

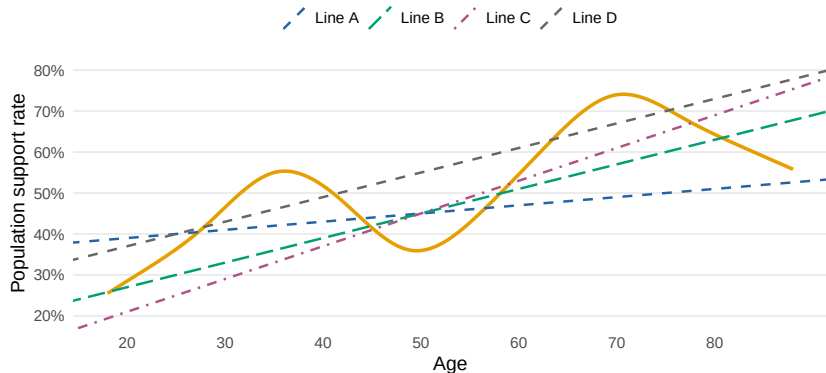
$$g(X) = a + bX.$$

- a is the intercept and b is the slope.
- This form is less flexible, but easier to summarize and interpret.
- We do not assume that the CEF itself is linear.

We call this the **linear regression line**.¹

¹Regression is a statistical method used to estimate the relationship between a dependent outcome variable and one or more independent predictor variables

Approximating the CEF with a Straight Line



The orange curve is the CEF. Each straight line is a candidate **linear predictor**, $a + bX$.

Different choices of a and b give different predictions.

Which line should we choose, and what makes it the best?

2. Best Linear Predictor

Comparing Linear Predictors Using MSE

Recall that the **mean squared error (MSE)** measures the average squared difference between the estimated values and the true value.

Definition

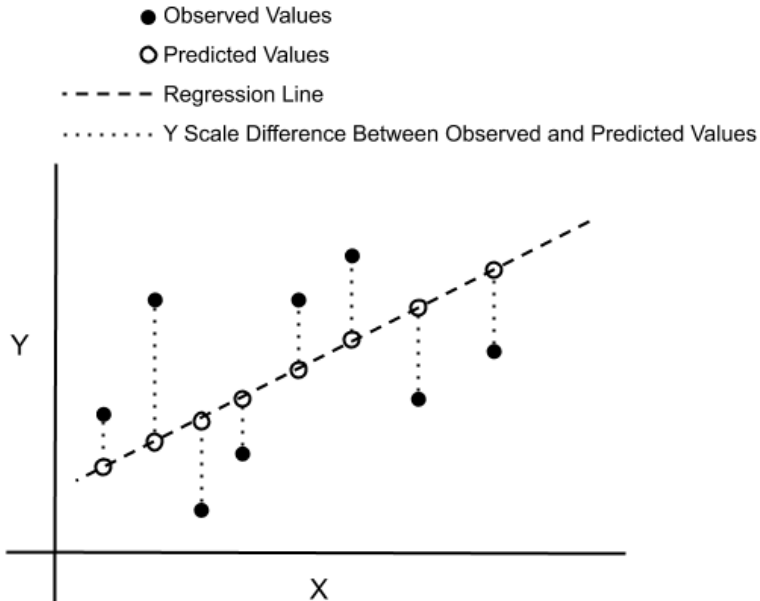
$$\text{MSE}(\hat{\theta}_n) = \mathbb{E} \left[(\hat{\theta}_n - \theta)^2 \right]$$

For a candidate linear predictor $a + bX$,

$$\underbrace{Y - (a + bX)}_{\text{Prediction error}} \quad \Longrightarrow \quad \underbrace{E \left[(Y - a - bX)^2 \right]}_{\text{Population MSE}}.$$

- Y : true value
- $\hat{Y} = (a + bX)$: predicted/estimated value

Mean Squared Error (MSE)



Intuition: Choosing the best linear predictor

Different choices of a and b produce different MSEs.

Our criterion

Choose the intercept a and slope b that give the **smallest population MSE**.

This choice gives us the **Best Linear Predictor**.

The Best Linear Predictor

Definition

The **Best Linear Predictor (BLP)** of Y given X is the linear function $\alpha + \beta X$ that minimizes the population mean squared prediction error:

$$(\alpha, \beta) = \arg \min_{a, b} E[(Y - a - bX)^2].$$

For any candidate line $a + bX$:

- $Y - (a + bX)$ is the prediction error.
- Squaring the error penalizes larger mistakes more heavily.
- Taking the expectation averages these squared errors over the population.

“Best” means **the smallest mean squared prediction error among all linear predictors**.

The CEF may be nonlinear; the BLP summarizes it with a straight line.

How Do We Find the BLP?

Choose the intercept and slope to minimize population mean squared prediction error:

$$Q(a, b) = E[(Y - a - bX)^2], \quad (\alpha, \beta) = \arg \min_{a, b} Q(a, b).$$

1. Differentiate

With respect to a :

$$\begin{aligned} \frac{\partial Q}{\partial a} &= E[2(Y - a - bX)(-1)] \\ &= -2E[Y - a - bX]. \end{aligned}$$

With respect to b :

$$\begin{aligned} \frac{\partial Q}{\partial b} &= E[2(Y - a - bX)(-X)] \\ &= -2E[X(Y - a - bX)]. \end{aligned}$$

Set both derivatives to zero at (α, β) :

$$\begin{aligned} E[Y - \alpha - \beta X] &= 0, & (1) \\ E[X(Y - \alpha - \beta X)] &= 0. & (2) \end{aligned}$$

2. Solve for the intercept

Expand condition (1):

$$E[Y] - \alpha - \beta E[X] = 0.$$

Rearrange:

$$\boxed{\alpha = E[Y] - \beta E[X].}$$

Next, expand condition (2):

$$E[XY] - \alpha E[X] - \beta E[X^2] = 0.$$

Substitute $\alpha = E[Y] - \beta E[X]$:

$$\begin{aligned} 0 &= E[XY] \\ &\quad - (E[Y] - \beta E[X])E[X] \\ &\quad - \beta E[X^2]. \end{aligned}$$

3. Substitute and solve for the slope

Expand and collect terms:

$$\begin{aligned} 0 &= E[XY] - E[X]E[Y] \\ &\quad + \beta(E[X])^2 - \beta E[X^2] \\ &= \underbrace{E[XY] - E[X]E[Y]}_{\text{Cov}(X, Y)} \\ &\quad - \beta \underbrace{(E[X^2] - (E[X])^2)}_{\text{Var}(X)}. \end{aligned}$$

Therefore,

$$\beta \text{Var}(X) = \text{Cov}(X, Y).$$

Divide by $\text{Var}(X) > 0$:

$$\boxed{\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}}.$$

The result: $\text{BLP}(X) = \alpha + \beta X$

Intercept: $\alpha = E[Y] - \beta E[X]$

The line's predicted value at $X = 0$.

It makes the line pass through $(E[X], E[Y])$.

Slope: $\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$

A one-unit increase in X changes the linear prediction by β units.

Best Linear Predictor

Theorem

For random variables X and Y with finite second moments and $\text{Var}(X) > 0$, the **Best Linear Predictor** of Y given X is

$$g(X) = \alpha + \beta X,$$

where

$$\alpha = E[Y] - \beta E[X], \quad \beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Corollary

If the CEF is linear, then the CEF and the BLP coincide:

$$E[Y | X] = \alpha + \beta X.$$

Linear Approximations

What if the CEF is **not linear**?

The BLP remains well-defined. It is the **linear projection** of Y onto a constant and X :

$$g(X) = \alpha + \beta X.$$

The CEF and the BLP solve different prediction problems:

- **CEF**: minimizes population MSE among **all functions of X** .
- **BLP**: minimizes population MSE among **linear functions of X** .

Key takeaway

The BLP does not require the CEF to be linear. It provides the best linear approximation even when the underlying conditional mean is nonlinear.

3. Ordinary Least Squares (OLS) estimator

From BLP to OLS

The **BLP** chooses the line that minimizes population MSE:

$$(\alpha, \beta) = \arg \min_{a, b} E[(Y - a - bX)^2].$$

In practice, we observe a sample rather than the entire population. We replace the population expectation with a sample average:

$$E[(Y - a - bX)^2] \longrightarrow \frac{1}{n} \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

Ordinary Least Squares (OLS)

OLS estimates the BLP coefficients by minimizing the sample mean squared prediction error:

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{a, b} \frac{1}{n} \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

BLP and OLS: What Does Each Term Represent?

Population: BLP

$$Y_i = \underbrace{\alpha + \beta X_i}_{\text{Population prediction}} + u_i$$

- Y_i : outcome for individual i .
- X_i : predictor for individual i .
- α : population intercept.
- β : population slope.
- u_i : prediction error relative to the BLP.

Sample: OLS

$$Y_i = \underbrace{\hat{\alpha} + \hat{\beta} X_i}_{\text{Fitted value } \hat{Y}_i} + \hat{u}_i$$

- Y_i : the same observed outcome.
- X_i : the same observed predictor.
- $\hat{\alpha}$: estimated intercept.
- $\hat{\beta}$: estimated slope.
- $\hat{u}_i$: residual relative to the fitted line.

OLS uses sample data to estimate the population BLP coefficients.

OLS: minimizes the sum of squared residuals (SSR)?

OLS replaces the population expectation with a sample average:

$$\underbrace{E[(Y - a - bX)^2]}_{\text{Population MSE}} \longrightarrow \underbrace{\frac{1}{n} \sum_{i=1}^n (Y_i - a - bX_i)^2}_{\text{Sample mean squared residuals}}.$$

For a given sample, n is a fixed positive number. Therefore,

$$\arg \min_{a,b} \underbrace{\frac{1}{n} \sum_{i=1}^n (Y_i - a - bX_i)^2}_{\text{Mean squared residuals}} = \arg \min_{a,b} \underbrace{\sum_{i=1}^n (Y_i - a - bX_i)^2}_{\text{Sum of squared residuals}}.$$

Same minimizing line

Minimizing the sum gives the exact same optimal slope and intercept parameters as minimizing the mean squared error (MSE), because dividing by a constant sample size n does not change the location of the minimum

OLS Is the Sample Analog of the BLP

Population BLP

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\alpha = E[Y] - \beta E[X]$$

Sample OLS

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

Plug-in principle: **Replace population moments with sample moments.**

Under random sampling and $\text{Var}(X) > 0$, the Law of Large Numbers implies:

$$\hat{\beta} \xrightarrow{P} \beta, \quad \hat{\alpha} \xrightarrow{P} \alpha.$$

OLS consistently estimates the BLP coefficients, even when the CEF is nonlinear.

The OLS slope has a closed form in terms of sample deviations

OLS Slope and Intercept

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}, \quad \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}.$$

- Numerator: sample covariation of X and Y
- Denominator: sample variation of X
- $\hat{\beta} = 0$ iff X and Y are sample-uncorrelated

Estimand, Estimator, and Estimate

Estimand / Parameter: What we want to learn

The population BLP slope:

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

A fixed, typically unknown feature of the population.

Estimator: The rule we use

The OLS slope estimator:

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

A function of the sample; its value varies across samples.

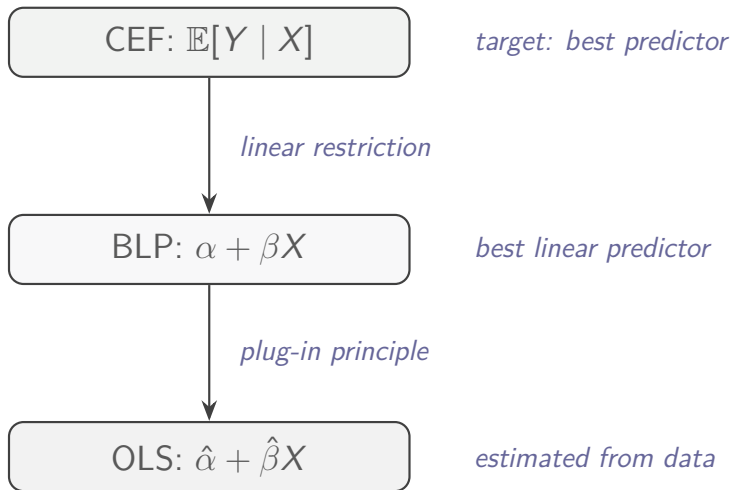
Estimate: The number we obtain

For example, applying OLS to an observed sample gives

$$\hat{\beta} = 0.006.$$

This numerical result is the estimate.

CEF, BLP, and OLS form a complete chain from target to estimator



Calculating OLS: By Hand and with lm()

```
#install.packages("wooldridge")
library(wooldridge)
data(wage1)

# OLS by hand: sample covariance / sample variance
beta_hat <- cov(wage1$educ, wage1$wage) / var(wage1$educ)
alpha_hat <- mean(wage1$wage) - beta_hat * mean(wage1$educ)

# OLS via lm()
#In `wage ~ educ`, `wage` is the outcome ($Y$) and `educ` is
#the predictor ($X$). It is read as "regress wage on educ".
ols <- lm(wage ~ educ, data = wage1)

# Compare the two methods
rbind(
  "By hand" = c(Intercept = alpha_hat, Slope = beta_hat),
  "Using lm()" = unname(coef(ols))
)
```

```
##           Intercept      Slope
## By hand    -0.9048516  0.5413593
## Using lm() -0.9048516  0.5413593
```

Understanding the OLS Output

```
summary(ols)
```

```
##
## Call:
## lm(formula = wage ~ educ, data = wage1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3396 -2.1501 -0.9674  1.1921 16.6085
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.90485     0.68497  -1.321   0.187
## educ         0.54136     0.05325  10.167 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.378 on 524 degrees of freedom
## Multiple R-squared:  0.1648, Adjusted R-squared:  0.1632
## F-statistic: 103.4 on 1 and 524 DF, p-value: < 2.2e-16
```

Our next goal: learn how to read and interpret the regression output.